

$$\begin{aligned} f_{obj} &= \sum_i w_i \{\nabla \cdot \hat{u}_t\}_i^2 \\ &= \sum_i w_i \left\{ \nabla \cdot \left(u_t^* - \frac{1}{\rho} \nabla \hat{p}_t \right) \right\}_i^2 \end{aligned}$$